MIL-P-70506A (AR)
4 April 1988
SUPERSEDING
MIL-P-70506 (AR)
04 December 1986

MILITARY SPECIFICATION

PROJECTILE, 155MM: XM864 BASE BURNER ASSEMBLY LOADING, ASSEMBLING AND PACKING

This specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for one type of projectile base burner assembly designated as Projectile, 155MM: XM864.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-286 - Propellants, Solid; Sampling; Examination and Testing

MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification

MIL-STD-453 - Inspection, Radiographic

MIL-STD-1234 - Pyrotechnics: Sampling, Inspection, and Testing

MIL-STD-1235 - Single and Multi-Level Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKING DRAWINGS

9381130 - Base Burner Assembly

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Proving ground tests. The base burner assembly shall comply with the following requirements when tested in accordance with 4.4.3.1.

3.2.1 Functioning. When assembled to projectiles the base burner assembly shall ignite, burn and enable the projectile to achieve its target range.

3.2.1.1 Horizontal range. No projectile shall have a horizontal range less than 97.5 percent of the mean horizontal range. Additionally, the range probable error (RPE) of the ballistic samples shall not exceed .441 percent when fired as specified in 4.5.8.

3.2.2 Security. There shall be no premature burst or indication of metal parts separation in the gun bore or in flight.

3.3. Igniter requirements.

3.3.1 Loose igniter composition. After consolidation there shall be no visual evidence of loose or flaking igniter composition. Any excess material (flash) shall be removed.

3.3.2 Heat of explosion. When tested in accordance with 4.5.2 the caloric output of the igniter material shall be 1150 $\pm$ 50 calories per grain.

3.3.3 Burn time test. When tested in accordance with 4.5.3 the igniter shall burn completely and provide a flame for at least 2.0 seconds.

3.3.4 Radiographic examination of igniters. When radiographed in accordance with 4.5.9 defects shall not exceed the following limits:

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a. The combined length of all cracks or laminations shall not exceed 0.20 inches. The width of any crack shall not exceed 0.02 inches.

b. The combined maximum length of all voids shall not exceed 0.10 inches. No single void shall have a dimension exceeding 0.05 inches.

c. The combined maximum length of all foreign matter shall not exceed 0.20 inches. No single particle of foreign material shall have a maximum dimension exceeding 0.10 inches.

d. There shall be no evidence of a gap between the bottom of the cup and the igniter material in excess of 0.01 inch.

3.4 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Records of results of contractor conducted examinations and tests. The contractor shall maintain records of the results of all examinations or tests required by this specification. (See 6.5).

3.6 Tool control dimensions. Dimensions marked tool control shall be gaged during First Article Testing and after eight hours maximum production on each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor. In the case of pressed igniters a correlation between punch dimensions and dimensions of the igniter surface need not be established. Inspection of the punch alone is adequate.

3.7 Radiographic examination of base burner assemblies. Sampled assemblies shall be radiographed and examined for the following:

a. Missing components.
b. Misassembled components.
c. Igniters facing the wrong way.
d. Foreign matter present.

3.8 Workmanship. All parts and assemblies shall be fabricated, loaded and assemblied in a thorough, workmanlike

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manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches that do not expose base material or remove the protective coating, are permitted. All required marking and stamping shall be neat and sharply defined. Sealants, lubricants and adhesives shall be utilized in accordance with the appropriate drawings.

3.8.1 Igniter cups. In addition to the above the igniter cups shall be free of holes, seams, scale, laminations, porosity, inclusions or imbedded foreign matter.

3.8.2 Igniter cup assembly. The stepped area of the pressed composition need not be a perfect form. Damaged areas up to 25 percent of the total surface area are allowed. Damage in excess of 25 percent is cause for rejection.

3.8.3 Marking. Marking on completed base burner assemblies shall be in accordance with the applicable drawing. The contractor shall develop and maintain a processing procedure which assures correct designation of the propellant lot and batch number on the Base Burner Assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements and for the provisions of MIL-A-48078 as specified herein.

4.1.1 Responsibility for compliance. All items must meet all requiremnts of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the quantities as indicated.

Part description	Drawings	Quantity
Base burner assemblies	9381130	12
Igniter	12630572	40
Igniter Composition	12630571	2 ounces
Weather seals	9390991	10
Retainer Assembly to Base prior to assembly of igniter housing	9381130	5
Igniter Housing Assembly	9381130	5

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70506A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Projectile, 155MM,: XM864 Base Burner Assembly Loading, Assembling & Packing				NEXT HIGHER ASSEMBLY See Below
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Igniter Composition (Dwg 12630571)				
	Heat of explosion	5	-		3.3.2, 4.5.2
	Total volatiles	1	-	3.1	4.5.1
	Igniter (Dwg 12630572)				
	Radiographic Examination	40			3.3.4, 4.5.9
	Examination for Defects	40	-	3.1	4.4.2.2
	Burn time test	30		3.3	4.5.3
	Base Burner Assembly (Dwg 9381130)				
	Examination for Defects	12	-	3.1	4.4.2.9
	Ballistic tests	10	-	3.2	4.5.8
	Radiographic examination	12	-	3.7	4.5.7

NOTES:

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70506A(AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
	Projectile, 155mm: XM864 Base Burner Assembly Loading, Assembling & Packing		2 2		NEXT HIGHER ASSEMBLY See below
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Retainer Assembly to base prior to assembly of igniter housing (Dwg 9381130) Examination for defects	5	-	3.1	4.4.2.4
	Igniter Housing Assembly (Dwg 9381130) Examination for Defects	5	-	3.1	4.4.2.3
	Weather Seals (Dwg 9390991) Examination for defects	10	-	3.1	4.4.2.7

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4.3.2 Inspection to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078. In addition, the first article sample shall be rejected if the ballistic test fails to comply with 3.2.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain:

a. Igniter housings from one lot interfix number from one manufacturer.

b. Igniters from one lot interfix number from one manufacturer.

c. Retainers from one lot interfix number from one manufacturer.

d. Grains from one lot interfix number from one manufacturer and not more than two lots.

e. Bases and base closures from one lot interfix number from one manufacturer and not more than two lots.

4.4.2 Examination. See MIL-A-48078 (AR).

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105 or MIL-STD-1235.

b. Automatic gage. Facilities equipped with Government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capabilty in accordance with the contract.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 12630571
4.4.2.1	Igniter Composition		NEXT HIGHER ASSEMBLY 12630572

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Heat explosion	*		3.3.2	4.5.2
102	Total volatiles	*		3.1	4.5.1
Minor	None defined				

NOTES:

* Sample size as specified in applicable test method paragraph

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70506A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Igniter Cup, Loading Assembly	1 OF 1	12630572
			NEXT HIGHER ASSEMBLY 9381104

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	Evidence of Loose, flaking or excess igniter composition		100%	3.3.1	Visual
Major					
101	Radiographic Examination		100%	3.3.4	4.5.9
102	Burn time test	*		3.3.3	4.5.3/4.4.3.2
103	Charge weight	*		3.1	4.5.5
104	Charge consolidation load and dwell time	*		3.1	4.5.4
105	Outside diameter, max		0.40%	3.1	Gage
106	Total length, max		0.40%	3.1	Gage
107	Depth from one edge to start of igniter, max		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship (Note 1)		0.65%	3.8	Visual

NOTES: *Sample size as specified in applicable test method paragraph

1. Provided the base metal is not gouged or cracked, damaged or missing plating is not cause for rejection.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Igniter housing assembly		9381130
			NEXT HIGHER ASSEMBLY: None

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Igniter cup missing or incorrectly oriented (Note 1)	1	0.40%	3.1	Visual
102	Plug missing		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:
1. To be inspected before assembly of plug.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70506A(AR)

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.4	Retainer to base prior to assembly of igniter housing	1 1	9381130
			NEXT HIGHER ASSEMBLY: None

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Excessive gap between base and retainer (360°)		0.40%	3.1	Gage
102	Proper igniter (color) correctly oriented (Note 1)		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:
1. To be inspected after assembly of retainer

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70506A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Igniter housing assembly to base prior to inserting grain and base closure	1 OF 1	9381130
			NEXT HIGHER ASSEMBLY: None

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Excessive gap between igniter housing and base cavity (360°)		0.40%	3.1	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.6	Grain to base assembly prior to insertion of base closure	1	1	9381130
				NEXT HIGHER ASSEMBLY: None

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Grain section missing		100%	3.1	Visual
b	Grain inhibitor perforated or cut		100%	3.1	Visual
Major 101	O-rings (2) present		0.40%	3.1	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70506A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Weather Seal		9390991
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%	3.1	Gage
102	Thickness		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70506A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Base Closure Prior to Assembly to Base				9381130
					NEXT HIGHER ASSEMBLY: None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Interior weather seal missing or punctured		100%	3.1	Visual
102	Exterior weather seal missing or punctured		100%	3.1	Visual
103	Location of interior weather seal		0.40%	3.1	Gage
104	Location of exterior weather seal		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Base burner assembly	1 OF 1	9381130
			NEXT HIGHER ASSEMBLY: None

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Excessive gap between base and base closure (360°)		100%	3.1	Gage
b	Air Pressure		100%	3.1	4.5.6
c	Ballistic Test	12		3.2	4.4.3.1, 4.5.8
Major					
101	Radiographic inspection See Note	1		3.1	4.5.7
102	Base threads damaged		0.40%	3.1	Visual
Minor					
201	Marking missing, misleading or unidentifiable including propellant lot and batch identification		0.65%	3.1	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

1. 50 per lot after qualification in accordance with 4.5.7.2.

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4.4.3 Testing. Testing is required as described herein and in the First Article and Quality Conformance Inspection Table.

4.4.3.1 Proving ground. These tests shall be performed as specified in 4.5.8. Defects shall be classified as follows:

Critical.

1. Base burner assembly failure inbore.

Special.

a. Range less than 97.5 percent of mean horizontal range.
b. Burner assembly fails to ignite

Major
101 Range probable error exceeds allowable limit

Minor: None defined.

4.4.3.2 Burn time test. A minimum of ten igniters shall be selected from the completed igniters of each batch of igniter material. If more than one consolidation machine is used then at least three igniters shall be selected from each machine. For example if four consolidation machines are used at least 12 igniters will be tested from each batch of material. Igniters are to be radiographed prior to burn testing.

4.4.4 Inspection equipment. For the inspection equipment required to perform the examinations and tests prescribed herein, the contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.4.4.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification or calibration check during production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by the above paragraph.

4.4.4.2 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment shall be qualified and certified in accordance with MIL-STD-410.

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4.5 Methods of inspection.

4.5.1 Igniter composition total volatiles. The contractor shall provide adequate controls to assure the compliance of each batch with the specified total volatiles requirement. For verification, a sample of each igniter composition from each 8 hours production and from each new batch shall be analyzed for total volatiles. If any sample fails to meet the requirements of the applicable drawing the production represented by the sample of material shall be rejected. The sample shall be tested in accordance with Method 102.1 of MIL-STD-1234.

4.5.2 Heat of explosion (25 atmospheres inert gas). The igniter material shall be tested for heat of combustion in a closed bomb calorimeter in accordance with Method 802.1 of MIL-STD-286. Two samples shall be taken from each batch of material produced. If both samples fail to yield a calorie per gram value in the specified range, (see 3.3.2); reject the material batch. If one sample fails, take a second double sample. If one or both samples fail from the second double sample then reject the batch. Accept a batch only when both samples of a double sample meet the requirement.

4.5.3 Burn time test. Complete igniter assemblies selected in accordance with 4.4.3.2 shall be tested to determine duration of flame. Suitable squib or electric match placed 1/4 to 1/2 inch from the open end of the cup shall be used to initiate the igniter material. Once lit, the igniter must produce a flame for at least two seconds. If the burn time is short, the igniters represented by the test sample shall be rejected. If after three attempts an igniter can not be initiated, the igniter and the group it represents shall be rejected. The burn time shall be recorded by use of a stopwatch or an electric eye. This test shall be performed at ambient temperature and pressure and must be done in a well ventilated area.

4.5.4 Igniter consolidation load. The contractor shall provide adequate controls to assure the compliance of each igniter with the specified load requirement and dwell time. For verification each consolidation machine shall be calibrated for consolidation load and dwell time at the beginning of each production shift or portion thereof using a load cell and stopwatch. In addition, a suitable warning device shall be assembled to each consolidation machine and calibrated at the same time as the consolidation machine. If, during the loading of the igniter the pressure drops before the applicable consolidation load requirement or the dwell time is short of the drawing requirement, that igniter shall be rejected.

4.5.5 Igniter charge weight. The contractor shall provide adequate controls to assure the compliance of each igniter with

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the specified charge weight requirement. At least once per hour for each igniter loading press the contractor will verify the accuracy of the igniter charge weight. To accomplish this an igniter cup assembly (cup) shall be identified and weighed. This cup assembly shall then be inserted into the loading press to be filled and consolidated in the normal operating mode. This igniter will be weighed. The difference between the before and after weights is the charge weight. Failure of any igniter to comply with the applicable charge weight requirement will require rejection of all igniters loaded since the last acceptable test.

4.5.6 Air pressure test of base burner assembly. The base assembly shall be placed in one or more chambers (the base to base closure joints as well as the weather seals must be tested) and a pre-determined volume of air shall be delivered to produce the specified pressure between the inside of the chamber and the outside of the assembled base burner. After initiation of test, the pressure must achieve the required level and maintain that level for the specified time interval. Failure to meet either criteria shall be cause for rejection, that assembly shall be suitably identified and removed from the lot to prevent mixing with accepted assemblies.

All equipment used in this test shall be capable of performing the test accurately and consistently. Equipment shall be verified at the start of each shift and thereafter, at intervals of not more than four (4) hours of continuous operation. It shall also be calibrated prior to starting the operation after a layoff of three (3) hours or more. If it is found that the gage is out of calibration in such a direction that the test pressure was actually lower than that specified on the drawing, base assemblies tested since the last acceptable calibration shall be retested after the gage error has been corrected.

4.5.7 Radiographic inspection of base burner assembly. The sampled base burner assemblies shall be radiographically examined for evidence of missing or misassembled parts prior to forwarding to a load, assembly and pack facility. If necessary, more than one radiograph shall be taken to avoid detrimental parallax effects. In the event the radiograph(s) taken in this view prevent a suitable examination, the assemblies shall again be radiographed at a location rotated 90° from the original. Assemblies and radiographs shall be suitably identified to permit cross reference. Radiographs shall be maintained by the contractor until final acceptance of the assembly lot. Radiographs shall be maintained by the contractor for 5 years after final acceptance of the base burner assembly lot. To verify adequate radiographic sensitivity a correctly assembled base burner unit, a unit with both igniters missing and a unit with both igniters inverted shall be radiographed. The unit with igniters missing and the unit with

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inverted igniters shall be assembled hand tight without o-rings or adhesive. Using the radiographic parameters which will be used for production inspection the units with missing and inverted igniters shall be easily discernable from the correctly assembled unit. With each group of inspection samples the unit with missing igniters and the unit with inverted igniters shall be radiographed and viewed to assure adequate radiographic sensitivity.

4.5.7.1 Radiographic qualifications. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. The personnel performing this examination shall be qualified and certified, level 2, per MIL-STD-410. Radiographic procedures including sample radiographs of a correctly assembled unit, a unit missing igniters and a unit with inverted igniters shall be submitted for approval in accordance with 6.3.

4.5.7.2 Frequency of radiographic inspection. When production is first initiated radiographic inspection of base burner units shall be performed on every assembly produced (100% inspection). A manufacturer shall qualify for reduced frequency of radiographic inspection when 500 consecutively produced assemblies meet the requirements of 3.7. After successful qualification, 50 units shall be randomly selected from each lot. Reduced sampling may begin in the middle of a lot, 50 samples will be selected from the remaining assemblies of that lot.

4.5.7.3 Radiographic defects. If any defect per 3.7 is found it shall be reported to the contracting activity. The particular defective unit shall be rejected and removed from the lot. If a defect is found in the reduced sample the lot represented by the sample shall be rejected and 100% inspection shall resume until requalification in accordance with 4.5.7.2 is achieved.

4.5.8 Proving ground test. The base burner assemblies used for this test shall be forwarded to a load, assembly and pack facility for assembly onto live loaded projectiles and forwarded to a designated government proving ground for firing. This test shall be conducted in accordance with an applicable proving ground test procedure. All projectiles fired for this test will be assembled with a M557 point detonating fuze and a live self registration charge.

The projectiles fired in this test shall be fired in the M199 Cannon using a M203 series propelling charge. The projectile, fuze and propelling charge shall all be at 70 degrees fahrenheit. A constant gun QE will be maintained throughout the test. Pressure time traces will be made for each projectile fired. Smear photographs will be taken at 10 and 30 meters in front of the muzzle. Doppler radar (HAWK) will be used.

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An effort will be made to fire all projectiles for this test within a one hour time span. Observation shall be made to determine compliance with 3.2. Stop the test if surface wind speed exceeds 20 knots or if two consecutive wind speed readings differ by more than 15 knots or if the wind direction of two consecutive readings differ by more than 800 mils (does not apply if wind speed is less than 5 knots).

4.5.8.1 Weapon and component data. All weapon and ammunition components used in support of this test shall be identified by model and lot or serial numbers.

4.5.8.2 Data reporting. As a minimum the proving ground test report will include the following observable attribute data:

a. Premature functioning
b. Metal parts separation
c. Impact functioning (information only)
d. Round by round range.

4.5.8.3 Calculation of percent RPE.

$$\text{Percent RPE} = \frac{67.45}{X}\sqrt{\frac{\sum_{i=1}^{10}(X-X_i)^2}{9}}$$

Where: X = Average range of group tested
x_i = Individual projectile range

4.5.8.4 Sample size. Twelve base burner assemblies will be randomly selected from the 50 base burner assemblies randomly selected for radiographic inspection (4.4.2.9). Prior to radiographic inspection the twelve assemblies shall be numbered 1 through 12 using a low stress metal stamp approximately two inches from the bottom of the assembly. A larger number shall also be placed in the same area of the assembly using a suitable marker. Ten assemblies are to be fired and counted. The remaining two assemblies are available for use of the proving ground should any of the projectiles fired be considered invalid due to test conditions (Note: The required RPE assumes a ten round sample).

4.5.9 Radiographic inspection of igniter cup assemblies. Each igniter cup shall be radiographically examined for voids, foreign matter, cracks and base gaps. The radiographic beam shall be directed towards the circumference of the igniter cup. The cup must be positioned when radiographed such that the base is clearly visible with no parallax distorting the cups features. Igniters

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failing the requirements of 3.3.4 shall be rejected and removed from the lot. Radiographic personnel, equipment and procedures shall meet the requirements of 4.5.7.1. Radiographic procedures and sample radiographs shall be submitted for approval in accordance with 6.3.

5. PACKAGING

5.1 Shipping. When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

5.2 Packaging, packing and marking. Packaging, packing and marking of base burner assemblies shall be in accordance with 9387108.

6. NOTES

6.1 Intended use. The XM864 Base Burner Assemblies covered by this specification are intended for the Projectile, 155MM: XM864 DPICM.

6.2 Ordering data. See MIL-A-48078 (AR).

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078 (AR). Submit equipment designs as required to Commander, USA ARDEC ATTN: AMSMC-QAR-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Submission of ballistic test data and data cards. In addition to the normal distribution of records, when the projectile is procured by the U.S. Army Armament Material and Readiness Command, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to : Commander, USA ARDEC, ATTN: AMSMC-QAR-Q(D), Picatinny Arsenal, NJ 07806-5000.

6.5 Data requirements. Data item descriptions required for submission under this specification are as follows:

Paragraph	Data Requirement	Applicable D.I.
4.5.7	Quality Inspection Defect Report	DI-R-1721
3.5	Quality Inspection Test Demonstration and Evaluation Report	DI-R-1724
4.4.4	Inspection Equipment Approval	DI-R-1714

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The above data shall be forwarded in accordance with the requirements of the contract to Commander, USA ARDEC, ATTN: AMSMC-QAR-Q (D), Picatinny Arsenal, NJ 07806-5000.

6.6 Subject term (keyword) listing.

Ammunition
Artillery
Military Specification
Carrot
Munition
Projectile, 155mm, XM864, Base Burner Assembly

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
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Preparing activity
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(Project 1320-A942)

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